Shizhao Ye

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Education

ShanghaiTech University Sep 2023 - Jun 2027 (expected)
Bachelor of Engineering in Computer Science and Technology
GPA: 3.72/4.00; Major GPA: 3.83/4.00 | Inaugural AI Honors Program
Selected coursework: Machine Learning (A), Probability and Statistics (A), Program Analysis (A), Numerical Optimization (A-), Algorithms and Data Structures (A-).

University of California, Berkeley Jan 2026 - May 2026
Exchange Student, UC Berkeley Extension Concurrent Enrollment
Spring 2026 coursework: COMPSCI 162: Operating Systems and System Programming (A-); COMPSCI 168: Introduction to the Internet: Architecture and Protocols (A); COMPSCI 185: Deep Reinforcement Learning, Decision Making, and Control (B+).

Research Interests

Embodied AI and robotics, with a focus on robot perception, generative models for robotics, and robot learning; broader interests in multimodal representations, sim-to-real transfer, and multi-robot systems.

Manuscripts & Preprints

M3GD: Multi-Modal Multi-View Geometric Diffusion for Camera-LiDAR Novel View Synthesis
Yang Zhou, Jiuhong Xiao, **Shizhao Ye**, Long Quang, Carlos Nieto-Granda, Giuseppe Loianno.
Submitted to IEEE Transactions on Robotics (T-RO), 2026.
arXiv:2609.30056

HuGo: LLMs as Whole-Body Policy Code Designers for Humanoid Loco-Manipulation
Seoyeon Choi, **Shizhao Ye**, Nicholas Bui, Aayushi Shrivastava, Kanghyun Ryu, Dhruva Tirumala, Markus Wulfmeier, Negar Mehr.
Submitted to IEEE ICRA 2027.
arXiv:2609.30594 | Project Webpage

Research Experience

Agile Robotics and Perception Lab (ARPL), UC Berkeley May 2026 - Sep 2026
Research Intern | PI: Prof. Giuseppe Loianno | Ph.D. Mentor: Yang Zhou

M3GD: Multimodal Novel View Synthesis
- Reproduced representative novel view synthesis (NVS) baselines and prepared RGB/depth visualizations; analyzed appearance and geometry failure modes to interpret M3GD's results.
- Extended RGB-LiDAR representation alignment analysis to the GrandTour dataset across DA3 feature levels and LiDAR aggregation strategies, providing evidence for architectural design choices.
- Identified diagnostic cases with large source-target viewpoint gaps to illustrate where target-view LiDAR provides the strongest benefits.

Robust NVS and Real-World Adaptation
- Built synthetic motion-blur and lens-contamination experiments; investigated feature restoration and LoRA adaptation to improve NVS robustness under degraded observations.
- Analyzed sensitivity to blur direction and severity; found improved target-view sharpness with LoRA while identifying persistent limitations under severe degradation.
- Contributed to ongoing deployment research by studying distribution shift and limited usable frame pairs in real-world data; explored LoRA adaptation and synthetic data augmentation.

Research Experience (continued)

Intelligent Control (ICON) Lab, UC Berkeley May 2026 - Sep 2026
Research Intern | PI: Prof. Negar Mehr | Ph.D. Mentor: Seoyeon Choi
HuGo: LLM-Generated Policies for Humanoid Loco-Manipulation

- Trained high-level PPO policies in NVIDIA Isaac Sim for humanoid loco-manipulation, providing RL baselines for comparison with LLM-generated policy code.
- Designed and debugged staged dense rewards for Push Button, Push Box, and Lift Box tasks; analyzed approach, end-effector alignment, contact, and grasp stability failures.
- Conducted hardware deployment experiments and debugging with a Vicon motion-capture system, supporting real-world evaluation of humanoid policies.

As3R Lab, ShanghaiTech University Oct 2024 - Present
Undergraduate Research Intern | Advisor: Prof. Yuqi Chen

- Conducted a literature review on foundation models for robotics and physical safety; developed a taxonomy of threats, defenses, and evaluation benchmarks for an ongoing embodied AI safety survey.
- Delivered a technical seminar on reinforcement learning fundamentals and safety guardrails (Jul 2025).

Teaching Experience

Teaching Assistant, ShanghaiTech University Sep 2025 - Jan 2026
SI100B: Introduction to Information Science and Technology

- Designed programming assignments, held office hours, and led recitation sessions to support students with course concepts and programming problems.

Selected Projects

Multi-Armed Bandit Analysis Dec 2024 - Jan 2025
Probability and Statistics | Team Project

- Implemented ε-greedy, UCB, and Thompson sampling; compared regret and convergence behavior through Monte Carlo simulations and visualized exploration-exploitation trade-offs.

HPC Optimization for ChaCha Encryption and Hash Tree Mar 2025 - Apr 2025
Computer Architecture | Individual Project

- Achieved a 15× speedup using AVX-512 SIMD and OpenMP; optimized memory access with cache blocking and loop unrolling.

Technical Skills

Programming: Python, C/C++, Java; basic MATLAB.
Robot Learning & Perception: PyTorch, NVIDIA Isaac Sim, PPO, reward design, diffusion models, LoRA, RGB-LiDAR analysis, Vicon motion capture.
Development: Linux, Git, GDB, OpenMP, AVX-512.
English: TOEFL 109/120 (R30, L27, S23, W29); CET-6.

Honors & Awards

Member of the Inaugural AI Honors Program, ShanghaiTech 2024 - Present
Selected based on outstanding academic performance and research potential.

Outstanding Student, ShanghaiTech University 2024 - 2025
Top 3%-7%

Outstanding Student, ShanghaiTech University 2023 - 2024
Top 7%-15%

Outstanding Student Cadre; Outstanding Youth League Committee Staff 2023 - 2024

Leadership & Service

Resident Assistant & Mentor Group Leader, Kedao College Sep 2023 - Present
Supported student transition and academic integration; organized college activities.

Staff Member, University Youth League Committee Mar 2024 - Present
Organized campus career events, including the ShanghaiTech Resume Competition and career fairs.